In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

ALL INDIA TEST SERIES

FIITJEE

JEE(Main) - 2017

ANSWERS, HINTS & SOLUTIONS

CONCEPT RECAPITULATION TEST – III
(Main)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	D	B	A
2.	B	D	C
3.	B	B	A
4.	C	B	A
5.	D	D	A
6.	C	C	D
7.	C	C	D
8.	B	B	C
9.	B	C	C
10.	A	D	C
11.	D	D	A
12.	D	C	A
13.	D	B	B
14.	A	D	C
15.	B	D	C
16.	D	A	D
17.	B	C	C
18.	B	D	D
19.	C	D	A
20.	A	C	B
21.	B	C	A
22.	A	D	B
23.	B	C	C
24.	A	C	A
25.	B	A	C
26.	B	B	C
27.	A	C	A
28.	B	B	C
29.	C	A	B
30.	C	A	B

Physics

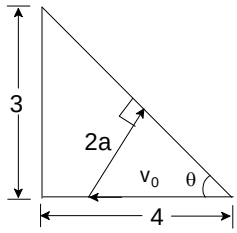
PART – I

SECTION – A

1. $\tan\theta = \frac{3}{4}$

$\sin\theta = \frac{2a}{v_0}$

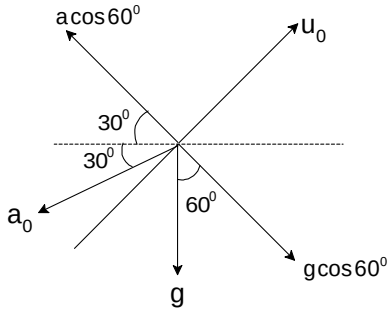
$\Rightarrow a = \frac{3}{2}$



2. From the frame of cabin

$\Rightarrow a_0\cos 60^0 = g\cos 60^0$

$\Rightarrow a_0 = 10\ m/s^2$



3. When 'v' is maximum

$g\sin\theta = kxg\cos\theta$

$\Rightarrow x = \frac{\tan\theta}{k} = \frac{3}{2}$

Also, $\left(g\sin\theta - kxg\cos\theta\right) = \frac{vdv}{dx}$

$\Rightarrow \frac{v^2}{2} = gx\sin\theta - \frac{kx^2}{2}g\cos\theta$ .....(i)

putting $x = \frac{3}{2}$, we get $v_{max} = 3m/s$

putting $v = \frac{3}{2}$ in equation (i)

we get x_1 & x_2 $(x_2 > x_1)$

$= \frac{3\sqrt{3}}{2}m$

4. $i_{rms}^2 \times R \times T = H$

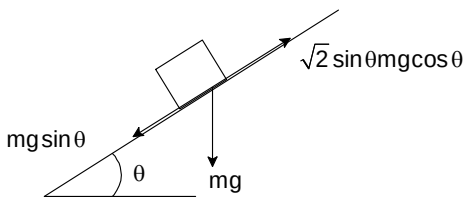
5. when sleeping starts for the first time

$\cos\theta = \frac{1}{\sqrt{2}}$

$\theta = \frac{\pi}{4}$

Just before sliding, f = $mg\sin\theta$

Just after sliding, f = $\frac{mg}{\sqrt{2}}\sin 2\theta$



6. % error = $\frac{\Delta V}{V}\times 100 = -6.25\%$

7. $ndt = m(v' + v_0)$

$(N\cos\theta - \mu N')dt = mv$

$N' = N\sin\theta$

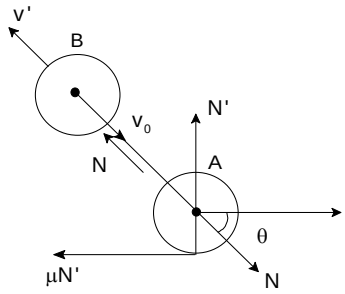
$\mu N'Rdt = \frac{2}{5}mR^2\omega$

for $\theta = 37^0$, we get $v = \frac{5v_0}{7}$

putting $v = R\omega$, we get $\tan\theta = \frac{4}{7}$

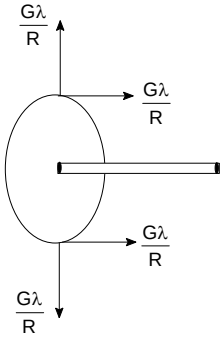
$\Rightarrow$ for $\theta > \tan^{-1}\left(\frac{4}{7}\right)$, friction will be static

& pure rolling will occur.



8. if $\Delta K = 0$, workdone by external force = ΔU

9. $\vec{F}_{net} = \frac{G\lambda}{R}m$



10. $\rho = \frac{\gamma P}{v^2}$

$\gamma = 1 + \frac{2}{F}$

$\rho = \frac{P}{v^2}.\left(1 + \frac{2}{f}\right)$

putting these value we get

$\rho = 1.3\ mg/cm^3$

11. $\frac{dv}{dT} = -\frac{v}{T(\gamma - 1)} \Rightarrow \gamma - 1 = \frac{v_0}{T_0\left(\frac{dv}{dT}\right)}$ at $v = v_0$, $T = T_0$ .....(i)

and $C_v = \frac{R}{\gamma - 1}$

$\frac{3R}{2} = \frac{R}{\gamma - 1}$ .....(ii)

putting the value of $\gamma - 1$

we get $\frac{dv}{dT} = \frac{3v_0}{2T_0}$

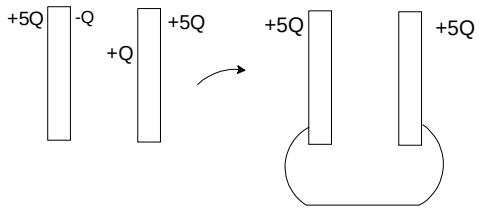
12. $I_C = 100 \times 0.04 \text{ mA} = 4 \text{ mA}$

$V_C = 20 - 12 = 8V$

$\Rightarrow R_C = \dfrac{8}{4 \times 10^{-3}} = 2000\Omega$

13. Heat $u_i - u_f$

$= \dfrac{1}{2}\dfrac{Q^2}{\left(\dfrac{\varepsilon_0 A}{d}\right)} - 0$



14. $W = \dfrac{1}{2}mv^2$

$= \dfrac{1}{2}\left(\dfrac{4}{3}\pi r^3 \rho\right)v^2 \qquad \left(V_{\text{terminal}} \propto r^2\right)$

$\Rightarrow W \propto r^7$

15. $\dfrac{1}{2}mv_0^2 = h\left(2f_0\right) - hf_0$

$\dfrac{1}{2}mv^2 = h\left(4f_0\right) - hf_0$

16. Let $v_x = v_y = v_z = v_0$

$\Rightarrow V_0 = \dfrac{V}{\sqrt{3}} \Rightarrow P = nm\left(\dfrac{2v}{\sqrt{3}}\right)$

17. $\dfrac{V^2}{R} = \dfrac{KAd\theta}{\ell}$

18. $\theta = \tan^{-1}\left(\dfrac{a}{g}\right)$

19. $\dfrac{KA(4-0)}{L-x} = \dfrac{3KA\left(0-(-2)\right)}{x}$

$\Rightarrow x = \dfrac{3L}{5}$

20. $\lambda_e = \dfrac{h}{m_e v_e}, E_e = \dfrac{1}{2}m_e v_e^2$

$\Rightarrow E_e = \dfrac{h^2}{2\lambda_e^2 me}$

$\Rightarrow E_p = \dfrac{hc}{\lambda_p} = \dfrac{hc}{n\lambda_e} = \dfrac{h\left(\dfrac{100nv_e}{2}\right)}{n\lambda_e}$

$\therefore \dfrac{E_e}{E_p} = \dfrac{1}{100}$

21. $(i)\ (\text{stress})_A = (\text{stress})_B$

$\Rightarrow \dfrac{T_A}{a_A} = \dfrac{T_B}{a_B}$ & $T_A x_1 = T_B(\ell - x_1)$

$\Rightarrow x_1 = \dfrac{2\ell}{3}$

$(ii)\ (\text{strain})_A = (\text{strain})_B$

$\Rightarrow \dfrac{Y_A}{\left(\frac{T_A}{a_A}\right)} = \dfrac{Y_B}{\left(\frac{T_B}{a_B}\right)}$ & $T_A x_2 = T_B(\ell - x_2)$

$\Rightarrow x_2 = \dfrac{6\ell}{7}$

Hence $x_1 : x_2 = 7 : 9$

$\begin{pmatrix} a \rightarrow \text{Area of cross-section} \\ T \rightarrow \text{Tension in the wire} \\ Y \rightarrow \text{Youngs Modulus} \end{pmatrix}$

22. $10q = \int_0^R 4\pi r^2 dr \rho_0 r$ $\qquad \int E.ds = \dfrac{q_{in}}{\epsilon_0}$

$\Rightarrow \rho_0 = \dfrac{10q}{\pi R^4}$ $\qquad \Rightarrow E(r) = \dfrac{1}{4\epsilon_0}\rho_0 r^2$

For '– 2q'

$\dfrac{(10q)}{\pi R^4}\dfrac{r_1^2}{4\epsilon_0}(2q) = \dfrac{1}{4\pi\epsilon_0}\dfrac{(2q)(4q)}{(r_1+r_2)^2}$ .....(i)

For '– 4q'

$\dfrac{(10q)}{\pi R^4}\dfrac{r_2^2}{4\epsilon_0}(4q) = \dfrac{1}{4\pi\epsilon_0}\dfrac{(2q)(4q)}{(r_1+r_2)^2}$ .....(i)

from (i) & (ii)

$r_2 = \dfrac{R}{\left(\sqrt{5}\left(\sqrt{2}+1\right)\right)^{\frac{1}{2}}}$

23. $\mu = \dfrac{\sin\left(\frac{A+D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}, D_m = A$

$\Rightarrow \cos\dfrac{A}{2} = \dfrac{1}{\sqrt{2}}$

$\Rightarrow A = 90^0$

24. For material with negative refractive index, ray bends or refracts differently as compared to materials with positive refractive index.

25. Use lens formula

26. $i^2 rdt = nC_p dT$

$i^2 = \frac{nC_p}{r}\frac{dT}{dt}$ $\qquad T = T_0 + \alpha t + \beta t^2 + \gamma t^3$

$i = \sqrt{\frac{2}{r}\frac{7R}{2}\left(\alpha + 2\beta t + 3\gamma t^2\right)}$ $\qquad \frac{dT}{dt} = \alpha + 2\beta t + 3\gamma t^2$

$i = \sqrt{\frac{7R}{r} \times 3}$

27. $A = \lambda N_0 e^{-\lambda t}$

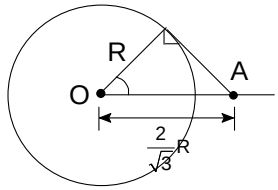
$\%\text{decrease} = \frac{A_0 - A}{A_0} \times 100$

$\Rightarrow \frac{1}{\lambda} = 20 \text{ days}$

28. $\theta = \frac{\pi}{6}$

$t = \frac{T}{6} = \frac{1}{6} \times \frac{2\pi}{\omega}$

$\Rightarrow \omega = \frac{\pi}{3t}$



29. $P = \frac{-B\partial s}{\partial x}$

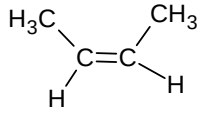
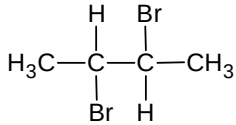
30. $\left(\frac{e}{a}\right)_{body} = E_{black\ body}$ for a particular wavelength.

Chemistry

PART – II

SECTION – A

1. because they must have same constitution and configuration at all other carbon atoms from C – 3 to C – 6 but different constitution at C – 1 & C – 2.

2. $H_3C{\equiv}CCH_3 \xrightarrow[\text{Lindlars catalyst}]{H_2}$  $\xrightarrow{Br}$  racemic

3. At cathode $= S\eta^{2+} + 2e^- \rightarrow S\eta$

   At anode $= Sn^{2+} \rightarrow Sn^{4+} + 2e^-$

   Total $S\eta$ lost from solution is 2×0.119 g of Sn^{2+}

   At cathode = so mass of reaming $SnCl_2 = 18.621$ g

   $$\& \text{ mass of } SnCl_4 = \frac{M_{SnCl_2}}{M_{Sn}} \times m_{Sn} = 0.26 \text{ g}$$

   $$\text{now } \frac{m(SnCl_2)}{m(SnCl_4)} = \frac{18.621}{0.260} = 71.62$$

4. ^-I & ^-M effect increases the nucleophilic reactivity

5. Buta – 1,3 – diene + styrene $\rightarrow$ Buna – s

6. It is curve of 2s so $(n + \ell) = 2 + 0 = 2$

7. $N = N_0\left(\frac{1}{2}\right)^n$ & no. of $t_{\frac{1}{2}} = \frac{20}{10} = 2$

   So, $\frac{N}{N_0} = \frac{1}{4}$

   Now, no. of decayed moles = $1 - \frac{1}{4} = \frac{3}{4}$

   Total amt. of He = $3 \times \frac{3}{4} = \frac{9}{4}$ mole

   Vol. at STP = $22.4 \times \frac{9}{4} = 50.4$ lit

8. $K_{sp}(AgCl) = 1.6 \times 10^{-10} = [Ag^+][Cl^-]$ .....(i)

   $K_{sp}(CuCl) = 1 \times 10^{-6} = (1s')(1s')$

   $$10^{-6} = \frac{1}{100} \times S^2 \qquad S^{12} = 10^{-4}$$

   $$S' = 10^{-2} = [Cl^-]$$

   $$= 1 \times 10^{-2}$$

   $$= 10^{-3}$$

now put S' in eq(i)

$$1.6\times10^{-10} = \left[Ag^{+}\right]\left[10^{-3}\right] = 1.6\times10^{-7}$$

so $x = 7$ so $10 - x = 7$

9. Cubic, orthorhombic, Rhombohedral, Tetrahedral, hexagonal.

10. Octalhedral void present at the centre of cube and tetrahedral void is present at $\left(\frac{1}{4}\right)^{th}$ of the distance along each body diagonal.

11. $KI + AgNO_3 \longrightarrow KNO_3 + AgI\downarrow$

Total moles of solute ions after reaction,

$$(0.1+0.2)\times I = 0.6 \text{ moles}$$

$$[\text{solute}] = \frac{0.6}{4}$$

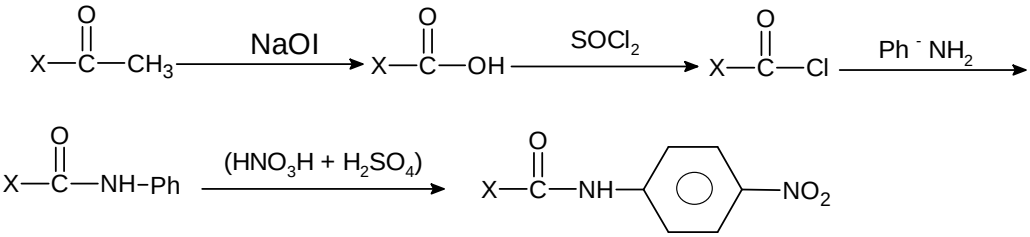
$$\Delta T_f = K_f \times \frac{0.6}{4} = 0.279 \text{ K}$$

12. $E_5 - E_1 = 96$

$$E_1 \times Z^2 \left\{\frac{1}{n_2^2} - \frac{1}{n_1^2}\right\} = 96$$

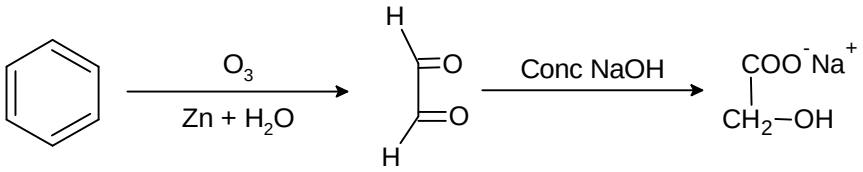
so $E_1 = 100J$

13. (B)



14. D

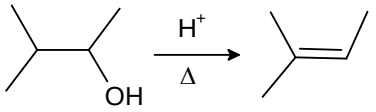
15.



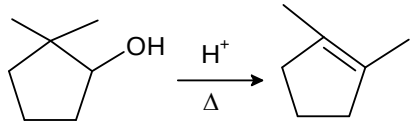
16. In basic solution Cr (III) converted Cr (VI) & it poses yellow chromate ion

17. Microcosmic salt is a white crystalline salt $Na(NH_4)HPO_4$ obtained by mixing solution of sodium phosphate & ammonium phosphate.

18. All rx^{n}s are involved in solvey process

19. 

$\alpha_H = 9 = P$

& 

$\alpha = 10 = Q$

so $P + Q = 19$

20. for equilibrium $\Rightarrow 2NO_2 \rightleftharpoons N_2O_4$

$6.8 = k_P = \dfrac{P'N_2O_4}{(P'NO_2)} = 6.8 \qquad \therefore P'N_2O_4 = 1.7$ atm

for IInd equilibriua

$$NO + NO_2 \rightleftharpoons N_2O_3$$

at $t = 0$ P 2P 0

at eq. $P - x$ $2P - x - 3.4$ x

since NO & NO_2 is in 1 : 2 ratio

at eq. $(P - x)$ 0.5 x

P'NO is same for both equilibria

now total $P = P - x + 0.5 + x + 1.7 = 5.05$

$x = 1.80$ atm

$P'NO = 1.05$ atm

21. due to max. force of attraction

22. $Av.Cv_1m = 2R$

For adiabatic process dv = dw

$$\frac{dr}{r} = -\frac{R}{C_{v'm}}\left(\frac{dv}{v}\right)$$

$\eta_1 C_{vm_1} dT + \eta_2 C_{vm2} dT$

$$= -(\eta_1 RT + \eta_2 RT)\frac{dv}{v}$$

$$\ell n\frac{T_2}{T_1} = -\frac{1}{2}\ell n\left(\frac{v_2}{v_1}\right)$$

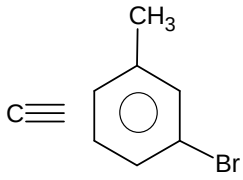
$$T_2 = 320 \times \left(\frac{1}{4}\right)^{\frac{1}{2}}$$

$= 160$k

$\Delta U = \left(\eta_1 C_{v_1 m_1} + \eta_2 C_{v_2 m_2}\right)\Delta T = -960$ R

23. Fe has one unpaired e^-

.

24. 

25. A (Fact based)

26. $2Cu_2S + 3O_2 \longrightarrow 2Cu_2O + 2SO_2 \uparrow$ does not occur in Bessemer's converter

27. X can't be Zn

28. Slowest step is R.d.s step

29. A is Na_2CrO_4 & B is H_2SO_4

30. Because Cr_2O_3 can't be reduced by C due to requirement of high energy.

Mathematics

PART – III

SECTION – A

1. A

The number of gaps when 10 persons seat is 11.

Let $x_1, x_2, x_3, \ldots, x_{11}$ be the number of vacants in the gaps

$x_1 + x_2 + ... + x_{11} = 30$

x_1 & $x_{11} \geq 0$ other $x_i \geq 2 \quad (i \neq 2, 11)$

Number of integral solution is $^{22}C_{10}$

2. C

Let the GP be a, ar, ar^2, …

$S_{2n} = \frac{a\left(1-r^{2n}\right)}{1-r}; S_n = \frac{a\left(1-r^{n}\right)}{1-r}; S_{3n} = \frac{a\left(1-r^{3n}\right)}{1-r}$

Clearly, $S_n, S_{2n} - S_n, S_{3n} - S_{2n}$ in GP

Let $S_n = A, S_{2n} - S_n = AR, S_{3n} - S_{2n} = AR^2$

Then, AR + A, 2AR, AR^2 + AR in HP as

$\frac{1}{A(R+1)} + \frac{1}{AR(R+1)} = \frac{1}{AR} = \frac{2}{2AR}$

3. A

We have $\frac{\alpha^2 + 3\alpha - 4}{2} = \frac{2\beta^2 - 3\beta - 2}{1} = K$

Clearly, $\alpha^2 + 3\alpha - 4 - 2K = 0$ has roots α & - 2β

$\therefore -2\alpha\beta = -(4 + 2K)$

$\Rightarrow \alpha\beta = 2 + K$

$\therefore \left|\frac{2 - \alpha\beta}{2K}\right| = \frac{1}{2}$

4. A

$\sum\left[(-1)^{q+r}\frac{n!}{p!q!r!}\sin^{2q}\theta.\cos^{2r}\theta\right]$, $n = p + q + r$, $x,p,q,r \in N$ is the expansion of

$\left(1 - \sin^2\theta - \cos^2\theta\right)^n = 0$

Hence no solution.

5. A

Contrapositive of $p \rightarrow \sim q \equiv q \rightarrow \sim p$

Inverse of $p \rightarrow \sim q \equiv q \rightarrow \sim p$

Now, inverse of $p \rightarrow \sim q \equiv \sim p \rightarrow q$

Contrapositive of $\sim p \rightarrow q \equiv \sim q \rightarrow p$

6. D

$0 < \frac{1}{2^x - 1} < 1$ and $0 < \ln|x| < 1$

$\Rightarrow 2^x - 1 > 1 \Rightarrow 0 < |x| < e$

$\Rightarrow x > 1$

$\therefore x \in (1, e)$

7. D

$\because (x^2 + 2x + 2) > 0 \ \forall x \in$

$\therefore \dfrac{mx^2 + 3x + 4}{x^2 + 2x + 2} < 5$

$\Rightarrow (m-5)x^2 - 7x - 6 < 0 \ \forall x \in R$

$m - 5 < 0 \ \& \ D < 0 \Rightarrow m < \dfrac{71}{24}$

8. C

$$\sum_{n=1}^{\infty}\left[\frac{1+2^n}{n!}(\ln 2)^n\right] = \sum_{n=1}^{\infty}\frac{(\ln 2)^n}{n!} + \sum_{n=1}^{\infty}\frac{(2\ln 2)^n}{n!}$$

$= e^{\ln 2} - 1 + e^{2\ln 2} - 1 = 4$

9. C

Point of concurrency (a + m, - 2)

$-2 = m(a+m) \Rightarrow m^2 + am + 2 = 0$

$a^2 \geq 8 \Rightarrow |a| \geq 2\sqrt{2}$

10. C

$$I = \int_0^4 \frac{1}{1+nf(x)}dx \Rightarrow I = \int_0^4 \frac{1}{1+nf(4-x)}$$

$$I + I = \int_0^4 \left[\frac{1}{1+nf(x)} + \frac{1}{1+nf(4-x)}\right]dx$$

$$= \int_0^4 \frac{2 + n(f(x) + f(4-x))}{1 + n^2 f(x) f(4-x) + n(f(x) + f(4-x))}dx$$

$$= \int_0^4 dx = 4$$

$\therefore I = 2$

11. A

Let A be (α, α) & $B(\beta, -\beta)$

P(h, k) be midpoint

$2h = \alpha + \beta$, $2k = \alpha - \beta$

$\alpha = h + k$; $\beta = (h - k)$

$\sqrt{(\alpha-\beta)^2 + (\alpha+\beta)^2} = d^2$

$\Rightarrow 4k^2 + 4h^2 = d^2$

Which is a circle

12. A

Given $a_{n-1}+1=\frac{a_n}{n}$

$\therefore \lim_{n\to\infty}\left(1+\frac{1}{a_1}\right)\left(1+\frac{1}{a_2}\right)...\left(1+\frac{1}{a_n}\right)$

$=\lim_{n\to\infty}\frac{a_{n+1}}{(n+1)!}=\lim_{n\to\infty}\left(\frac{1+a_n}{n!}\right)$

$=\lim_{n\to\infty}\left(\frac{1}{n!}+\frac{1}{(n-1)!}+\frac{a_{n-1}}{(n-1)!}\right)$

as $a_1=1$

Unit value = $\lim_{n\to\infty}\frac{1}{n!}+\frac{1}{(n-1)!}+...+\frac{1}{n}=e$

13. B

$f(x)=\lim_{t\to\infty}\left\{\frac{(1+\sin\pi x)^t-1}{(1+\sin\pi x)^t+1}\right\}$

$=\begin{cases}1, & 2n<x<2n+1\\ -1, & 2n+1<x<2n+2\\ 0, & x=0,1,2,.....\end{cases}$

14. C

$px^3+qx^2+r=0 \quad \alpha,\beta,\gamma$

$\alpha\beta+\beta\gamma+\gamma\alpha=0$

$\therefore \begin{vmatrix}\alpha\beta & \beta\gamma & \gamma\alpha\\ \beta\gamma & \gamma\alpha & \alpha\beta\\ \gamma\alpha & \alpha\beta & \beta\gamma\end{vmatrix}=\begin{vmatrix}\alpha\beta+\beta\gamma+\gamma\alpha & \beta\gamma & \gamma\alpha\\ \alpha\beta+\beta\gamma+\gamma\alpha & \gamma\alpha & \alpha\beta\\ \alpha\beta+\beta\gamma+\gamma\alpha & \alpha\beta & \beta\gamma\end{vmatrix}\left[C_1\to C_1+C_2+C_3\right]=0$

15. C

$A=\begin{vmatrix}0 & 1 & 2\\ 0 & 0 & 3\\ 0 & 3 & 1\end{vmatrix}$

Find A^2, A^3 to verify

16. D

$\vec{a}+\vec{b}=k\vec{c}$ ...(I)

$\vec{b}+\vec{c}=\ell\vec{a}$ ....(II)

(I) – (II)

$\vec{a}-\vec{c}=k\vec{c}-\ell\vec{a}$

$\Rightarrow(1+\ell)\vec{a}+(1+k)\vec{c}=0$

$\Rightarrow \ell=k=-1$

i.e $\vec{a}+\vec{b}+\vec{c}=0$

17. C
The points A, B and C are (3p, 0, 0); (0, 3q, 0) and (0, 0, 3r) respectively.

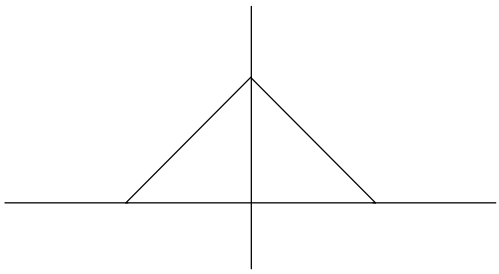
18. D
Agreement happens when both speak either the truth or both tell a lie
E_1 = A and B both speak truth
E_2 = A and B both tell a lie
E = A and B agree in a certain statement
$\therefore P\left(\frac{E_1}{E}\right) = \frac{xy}{1-x-y+2xy}$

19. A
$P(X = x) = K(x+1)\frac{1}{5^x}$, K is proportionality constant
$k = \frac{16}{25}$
Only (B) is true

20. B
$$\int \frac{1}{\sqrt{1+x^2}\left(\sqrt{1+x^2}-x\right)^{15}}dx = \int \frac{\left(\sqrt{1+x^2}+x\right)^{15}}{\sqrt{1+x^2}}dx$$
$$= \frac{1}{15}\left(\sqrt{1+x^2}+x\right)^{15} + c$$

21. A
$I = \int_0^1 \frac{dx}{1+x^{\frac{\pi}{2}}}$
In (0, 1)
$I > \int_0^1 \frac{dx}{1+x} \Rightarrow I > \ln 2$
And $I < \int_0^1 \frac{1}{1+x^2}dx \Rightarrow I < \frac{\pi}{4}$

22. B
$f(x) = \begin{cases} 1-|x|, & |x| \le 1 \\ 0, & |x| > 1 \end{cases}$



$f(x)$ is even function

$g(x) = f(x-1) + f(x+1)$

$g(-x) = f(-x-1) + f(-x+1)$

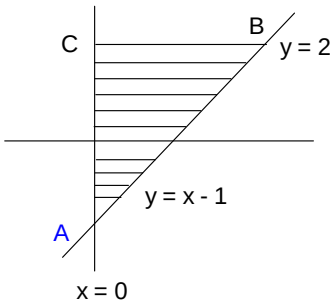
$= f(x+1) + f(x-1)$

$\therefore$ g(x) is also even only.

23. C

$y = \left[\frac{x^2}{64} + 2\right]$; $y = x - 1$ or $x = 0$

$y = \sin^{-1}\left[\frac{|x|}{4}\right] = 0$ for $|x| < 4$



$\therefore$ area $= \frac{1}{2} \times 3 \times 3 = \frac{9}{2}$

Area above x-axis = $\frac{9}{2} - \frac{1}{2} \times 1 \times 1$

= 4

24. A

$2y \sin x \frac{dy}{dx} + y^2 \cos x = \sin 2x$

$\Rightarrow \frac{d(y^2 \sin x)}{dx} = \sin 2x$

$\Rightarrow y^2 \sin x = -\frac{\cos 2x}{2} + c$

At $x = \frac{\pi}{2}$

$1 = \frac{1}{2} + c \Rightarrow c = \frac{1}{2}$

$y^2 \sin x = \frac{1 - \cos 2x}{x}$

$\Rightarrow y^2 = \sin x$

25. C

$\frac{df(\tan x)}{dg(\sec x)} = \frac{f'(\tan x)\sec^2 x}{g'(\sec x)\sec x.\tan x}$

At $\tan x = 1 \& \sec x = \sqrt{2}$

$x = 2n\pi + \frac{\pi}{4}, n \in N$

$\therefore \frac{df(\tan x)}{dg(\tan x)} = \frac{2\times 2}{4\times\sqrt{2}} = \frac{1}{\sqrt{2}}$

26. C

Let $\omega = \frac{z-1}{e^{i\theta}} + \frac{e^{i\theta}}{z-1}$

If $z = x + iy$

$\omega = \frac{(x-1)+iy}{\cos\theta + i\sin\theta} + \frac{\cos\theta + i\sin\theta}{(x-1)+iy}$

$Im(\omega) = 0$

$\Rightarrow -\sin\theta(x-1) + y\cos\theta + \frac{\sin\theta(x-1) - y\cos\theta}{(x-1)^2 + y^2} = 0$

$\Rightarrow \sin\theta\left[(x-1) + \frac{x-1}{(x-1)^2+y^2}\right] + \cos\theta\left[y - \frac{y}{(x-1)^2+y^2}\right] = 0$

$\Rightarrow (x-1)\left[(x-1)^2 + y^2 - 1\right] = 0$ and $y\left[(x-1)^2 + y^2 - 1\right] = 0$

$\therefore$ locus is circle with unity radius

27. A

$\tan\alpha.\tan 2\alpha + \tan 2\alpha.\tan 4\alpha + \tan 4\alpha.\tan\alpha$

$= \frac{\sin\alpha.\sin 2\alpha\cos 4\alpha + \sin 2\alpha.\sin 4\alpha.\cos\alpha + \sin 4\alpha.\sin\alpha.\cos 2\alpha}{\cos\alpha.\cos 2\alpha.\cos 4\alpha}$

$= \frac{\cos\alpha.\cos 2\alpha.\cos 4\alpha - \cos(\alpha + 2\alpha + 4\alpha)}{\cos\alpha.\cos 2\alpha.\cos 4\alpha} = 1 - \frac{8\sin\alpha}{\sin 8\alpha} = -7$

28. C

$\frac{1}{3}\left(x^2 - 12x + 39\right) = \sin\left(\frac{a}{x}\right) \Rightarrow \frac{(x-6)^2}{3} + 1 = \sin\left(\frac{a}{x}\right)$

$\Rightarrow \frac{a}{x} = 2n\pi + \frac{\pi}{2}$ & $\frac{x-6}{3} = 0 \Rightarrow x = 6$

$\therefore a = 3\pi(1 + 4n), n \in z$

29. B

Let the parabola be $y = Ax^2 + Bx + C$

(1, 1) satisfies $\Rightarrow A + B + C = 1$ .... (1)

$\left.\frac{dy}{dx}\right|_{x=1} = 1 \Rightarrow 2A + B = 1$ ....(2)

$f(0) = C$ & $f'(0) = B$

$\therefore 2f(0) + f'(0) = 1$

30. B

$\sum_{i=1}^{10}(x_i - 4) = 10 \Rightarrow \sum_{i=1}^{10}(x_i - 5) = 0$

Also, $\sum_{i=1}^{10}(x_i-4)^2=100 \Rightarrow \sum_{i=1}^{10}(x_i-5)^2=90$

$\sigma=\sqrt{\frac{1}{10}\times 90}=3$

$\therefore$ right triangle but scalene.